A cellular prior enables compact and transferable single-cell inference with CellPFN

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Abstract

Single-cell prediction often requires learning from limited labelled context across heterogeneous experiments. We developed CellPFN, a prior-data fitted network trained on a procedural cellular prior that varies biological signals, donor structure and measurement noise. The 12.33-million-parameter model uses task-local gene correspondence without persistent gene identities and requires no target-task gradient updates. After 384,000 scheduled synthetic episodes, CellPFN achieved competitive cell-type classification relative to pretrained tabular models, with approximately 90-fold less reported task exposure than TabICLv2. The cellular prior improved three pretrained tabular models through continued training, improving one-shot balanced accuracy by 5.6–10.8 percentage points. CellPFN also accepted transcriptome-scale inputs and supported efficient gene-deletion explanations. **The framework further identifies neuropathology-associated cellular states and reproducible predictive expression signatures across independent cohorts.**[O04] These results position a cellular task prior as a reusable resource for learning biological inference with limited labelled data.

Single-cell atlases provide increasingly broad reference maps, yet annotating a new experiment often requires inference from a small labelled set collected under different biological and technical conditions. More profiled cells do not necessarily supply more independent donors or more varied experimental settings. In supervised single-cell annotation, increasing donor diversity can contribute more to generalization than increasing cell count alone[1]. This distinction motivates models that learn how to interpret limited experimental context across heterogeneous tasks.

Single-cell foundation models such as Geneformer, scGPT, scFoundation, UCE and TranscriptFormer learn reusable representations from large collections of measured profiles[2–6]. Their breadth supports transfer across biological settings, but large-scale pretraining does not guarantee an advantage on each downstream task. Evaluations have found limitations in frozen representations and

saturation as pretraining collections expand[7,8]; recent preprint evidence also shows that normalization and linear methods can remain competitive across several benchmarks[9]. These findings motivate a complementary question: can learning improve the inference rule that combines a few labelled examples, even when the expression measurements themselves already provide useful features?

Meta-learning already supports transfer across heterogeneous single-cell experiments, as demonstrated by MARS[10]. Prior-data fitted networks (PFNs) provide a complementary way to learn inference rules. They are trained across prediction problems drawn from a generative prior, amortizing support-conditioned inference into a forward pass[11]. TabPFN and TabICL demonstrate that synthetic task distributions can produce effective general-purpose tabular learners[12–15]. Recent MapPFN and PerturbPFN work further applies prior-based learning to cellular perturbation prediction[16,17]. For transcriptomic classification, the opportunity is to encode the variation encountered across cellular experiments directly into the training prior, while accommodating thousands of sparse features and scarce labels.

Single-cell simulators already model count distributions, dependence, donor variation and regulatory dynamics through complementary statistical and mechanistic approaches[18–25]. Learned generative models also support conditional synthesis and augmentation[26,27]. For PFN training, the design objective extends beyond reproducing a reference dataset to generating diverse labelled inference problems. Controlled variation in signal strength, hidden subpopulations, donor effects and measurement noise can expose a learner to experimental configurations without requiring each configuration to be measured and annotated. Synthetic-to-real learning in optical flow and face analysis provides precedents for learning from known simulated structure[28,29]. Nevertheless, simulator fidelity depends on the statistic and scenario being evaluated[30,31]. The relevant test is whether the induced prior teaches inference that transfers to real biological data.

Here we introduce CellPFN and its procedural cellular prior. CellPFN learns from synthetic tasks while preserving gene correspondence within each task and omitting persistent gene identities across tasks. We test two consequences of this prior: whether a compact cellular model can achieve competitive biological inference after substantially fewer training episodes than a general tabular model, and whether tasks from the same simulator can improve released tabular models through continued training. We further examine transcriptome-scale inputs and efficient query-gene explanations. The aim is a reusable method for learning from limited labelled biological context, with the prior's usefulness established through downstream inference rather than simulation scale alone.

Results

A cellular prior teaches inference from labelled context

We constructed a procedural prior over single-cell prediction problems, varying biological differences together with the conditions under which they are observed. The generator samples hierarchical cell identities, distributed expression programs, rare markers, correlated variation and donor-dependent

states. Donor composition, study and batch shifts, sequencing depth and overdispersed count sampling produce heterogeneous observations of these latent signals. Different labels and support budgets turn each simulated dataset into multiple inference problems. This makes experimental variation part of the training distribution, with known task labels and controllable nuisance factors. Aggregate statistics from real datasets, including those later evaluated here, informed the prior's ranges. All examples used to optimize CellPFN's weights were synthetic.

Each problem comprises labelled support cells and unlabelled queries. Genes retain their correspondence between support and query but have no persistent identity across problems. CellPFN compares each query gene with support-derived class summaries, pools the resulting relations through permutation-invariant statistics and integrates this evidence into class scores. This use of unordered features follows the principle of set-based learning[32], while preserving the within-task correspondence required for expression comparisons. A covariance estimate fitted to the support cells supplies an additional statistical channel when support is sufficient. The pooled evidence produces direct neural class predictions. Prediction on a new task requires labelled context but no gradient updates, and queries do not condition on other query cells.

The reference model has 12.33 million trainable parameters and was trained over 384,000 scheduled episodes drawn from 12,000 simulated dataset banks. Reused banks supplied distinct task views; 276,828 episodes carried nonzero loss weight. The first 320,000 scheduled episodes used prior v1, followed by 64,000 using the revised v2.2-B prior. For scale, TabICLv2 reports approximately 35 million synthetic pretraining datasets[15], about 90 times CellPFN's scheduled episode count. This comparison describes training-task exposure across different priors and architectures; it does not measure a matched-compute advantage.

Compact CellPFN is competitive across biological inference regimes

We evaluated coarse and fine cell-type annotation in peripheral blood mononuclear cells (PBMCs)[33], together with cortex annotation within the development collection and on queries from held-out studies in ZEBRA[34]. Each panel used one, ten or 100 labelled support cells per class. Comparators included support-trained statistical classifiers, released tabular models, and logistic classifiers fitted to Geneformer and scGPT embeddings[2,3,15,35,36]. The main comparisons used the same three support draws. PBMC inputs were counts for CellPFN and normalized expression for tabular comparators; the cortex analysis used the exported processed-expression view. Model-specific input routes were retained for the single-cell foundation models; the Geneformer logistic classifier used standardized embeddings, whereas the scGPT classifier used unscaled embeddings.

Across the four panels and three support sizes, CellPFN using all available genes reached a mean balanced accuracy of 72.9%. Released TabICLv2 and Kumo large reached 66.8% and 70.1%, respectively, using 2,000 highly variable genes, while cellular continued training brought Kumo large to 72.4%. Geneformer and scGPT with logistic classifiers reached 74.9% and 69.4% through their native input routes. These descriptive averages place the compact cellular model in a competitive range without establishing statistical equivalence or a universal ranking.

Support-trained low-rank linear discriminant analysis (LDA) remained strong when labels were abundant. At 100 support cells per class, LDA using all genes reached 84.6% mean balanced accuracy across the four panels, compared with 85.4% for CellPFN; at ten supports, the corresponding values were 75.3% and 79.6%. LDA received processed expression throughout and was not evaluated with one support cell per class.

The ranking depended on the biological task and the amount of labelled context. For fine PBMC annotation with one support cell per class, Geneformer reached 68.8% balanced accuracy versus 51.1% for CellPFN. For cortex queries from held-out studies with ten support cells per class, CellPFN reached 91.9%, compared with 87.4% for Geneformer and 87.2% for scGPT. Both foundation models likely encountered the exact PBMC cells during unlabelled pretraining; Geneformer also encountered source studies contributing most held-out cortex queries. This exposure does not establish label leakage or explain the performance differences, which also reflect preprocessing and classifier settings. Moreover, model variants were selected on these development panels, and each training arm currently has one training seed. **A frozen evaluation on additional, independently held-out cohorts will determine how consistently this competitive performance generalizes.**[O01]

The cellular prior also improves pretrained tabular models

To test whether the prior's utility extends beyond CellPFN, we continued training released TabICLv2, Kumo small and Kumo large on tasks from the same simulator, using prior v2.2-B throughout cellular adaptation. Each model received 40,000 additional cellular tasks and 10,000 generic replay tasks over 2,500 updates. With 2,000 highly variable genes at evaluation, one-shot balanced accuracy improved by 7.3, 10.8 and 5.6 percentage points, respectively, averaged across the four core panels. Changes at 100 support cells per class were smaller than one percentage point in magnitude. Thus the largest benefits arose where labelled biological context was most limited.

A generic-table continued-training control for Kumo small changed mean balanced accuracy by approximately zero across the eight low-support panel–budget combinations, compared with a 6.1-point gain for cellular continued training. This comparison supports a contribution from the training distribution, although the controls differed in rows and columns per update and equivalent controls were unavailable for the other architectures. The experiments therefore demonstrate two uses of the cellular prior: training a compact specialist from scratch and adapting already pretrained tabular learners.

Gene-anonymous inference extends to transcriptome-scale inputs

We next asked whether CellPFN could use more genes than it encountered during training. The largest training tasks contained 8,192 genes, whereas the PBMC and processed cortex evaluations contained 33,538 and 21,158 genes, respectively. The same shared per-gene computations and symmetric aggregation accepted these larger inputs without expanding a learned gene vocabulary. Across the matched benchmark draws, balanced accuracy increased from 71.2% with 2,000 highly variable genes to 73.8% with 8,192, and was 72.9% with all genes. Additional genes therefore

remained usable, although retaining every gene did not maximize aggregate accuracy and was not beneficial in every task.

Released TabICLv2 declined from 66.8% to 48.3% and Kumo large from 70.1% to 66.4% when moving from 2,000 highly variable genes to all genes. TabPFN-3.5, evaluated under its non-commercial research terms, refused the all-gene cases at its default feature guard[36,37]. These outcomes describe the tested inference configurations, including their column handling; they do not establish a fundamental limit of tabular architectures. CellPFN's advantage here is the ability to retain transcriptome-scale input under its existing inference procedure.

Prior and architecture changes have regime-dependent effects

We examined whether explicitly exposing cross-gene structure accounted for CellPFN's performance. The support-fitted covariance channel supplied both per-gene variance information and low-rank relationships between genes. Later comparisons of separately trained models with no operator, diagonal information alone or the full operator showed gains in several support-rich settings. However, the prespecified operator test did not support its primary claim: an operator-attributable loss exceeded the allowed threshold on a primary task. The later descriptive improvements therefore do not establish the operator as uniformly beneficial, and variation across independent training runs remains unmeasured.

Changing the simulated task distribution also produced a tradeoff. Continuation on tasks selected as learnable improved some low-support accuracies but worsened predictive log loss at larger support sizes; the comparison additionally changed the number of cells per update. Simulator diagnostics likewise showed partial agreement with real data rather than uniform fidelity: the prior covered the observed strongest-marker range, while its few-cell discrimination tasks were easier than the real reference. Together, these results motivate judging the prior through transfer and targeted controls, alongside distributional checks, rather than treating greater simulated realism or a stronger statistical channel as sufficient evidence of better inference.

Query-gene deletion explains predictions efficiently

Gene anonymity still permits gene-level explanations because feature names can be restored after prediction. We developed DeepSets leave-one-gene-out (DS-LOO), which estimates how setting an individual query gene to zero changes the logit margin between the initially predicted class and its runner-up. The support set and fitted covariance remain fixed. In contrast to gradient attribution along an input path[38], the target is this finite query-side deletion. DS-LOO uses the pooled gene statistics and a linearization of the prediction head, with corrections for changes in depth and detected-gene fraction. It is an approximation to a model intervention, not a simulated biological gene knockout.

Against exact deletions in six evaluated cases and two model states, the case-level medians of within-cell rank correlation ranged from 0.979 to 0.998. For the reference model, case-level corrected relative L2 errors ranged from 0.5% to 4.9%. These checks used selected genes and correctly

classified cells in full precision. **Under matched load, computing all-gene explanations cost approximately 10–13 single-gene exact forward passes. For the PBMC example, DS-LOO took 67.9 seconds; the corresponding 61-hour exhaustive-deletion runtime was extrapolated from individual forward timings rather than measured directly. Shared-GPU timings remain provisional.**[O05]

Despite the absence of gene-name embeddings, IL7R ranked second for CD4 T cells and NKG7 seventh for natural killer cells in the evaluated class-margin rankings. Removing each PBMC query's 50 most influential genes reversed its margin against the original runner-up class in 45.0% of evaluated cells, compared with 0.3% for random expressed genes. This supports faithfulness to the classifier's decisions. It does not show that the same deletions would change biological identity or predict experimental perturbation responses.

A provisional application examines neuropathology-associated expression

We extended the framework to donor-level neuropathology contrasts in ROSMAP-derived single-nucleus data[39]. In the exported analysis with case and control support matched within sex, microglial predictions distinguished the CERAD 4-versus-1 donor groups with an AUC of 0.745 and a reported corrected 95% repeated-split interval of 0.700–0.790. This result remains provisional pending completion of the donor-label audit and a frozen rerun. It measures association with neuropathology, not prospective diagnosis or clinical utility.[O02]

Applying query-gene deletion to the corrected disease analysis and testing the findings in independent cohorts will test whether this approach can connect donor-level prediction to reproducible cellular expression signatures.[O03]

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